Question	Answer	Marks
1(a)	acceleration and displacement identified as vectors (and no others)	**B1**
	speed, temperature and gravitational potential energy identified as scalars (and no others)	**B1**
1(b)(i)	$W = Fs$ **or** $W = mas$	**B1**
	$s = v^2 / 2a$ **or** $a = v^2 / 2s$ **or** $as = v^2 / 2$	**B1**
	$W = ma(v^2 / 2a)$ **or** $W = m(v^2 / 2s)s$ **or** $W = m(v^2 / 2)$ **and** (so E_K)= $\frac{1}{2}mv^2$	**B1**
	OR $W = Fs$ or $W = mas$	**(B1)**
	$F = mv / t$ **and** $s = \frac{1}{2}vt$	**(B1)**
	$W = mv / t \times \frac{1}{2}vt$ **and** (so E_K)= $\frac{1}{2}mv^2$	**(B1)**
	OR $W = Fs$ or $W = mas$	**(B1)**
	$a = v / t$ **and** $s = \frac{1}{2}vt$	**(B1)**
	$W = m(v / t)(\frac{1}{2}vt)$ **and** (so E_K)= $\frac{1}{2}mv^2$	**(B1)**
	OR $W = Fs$ **or** $W = mas$	**(B1)**
	$a = v / t$ **and** $s = \frac{1}{2}at^2$	**(B1)**
	$W = m(v / t)(\frac{1}{2} \times (v / t) \times t^2)$ **and** (so E_K)= $\frac{1}{2}mv^2$	**(B1)**

Question	Answer	Marks
1(b)(ii)	kinetic energy $= \frac{1}{2} mv^2$ $= \frac{1}{2} \times 920 \times 17^2$ $= 1.3 \times 10^5$ J	**A1**
1(b)(iii)	$P = W/t$	**C1**
	$= (4.7 \times 10^4 + 1.3 \times 10^5) / 5.8$	**C1**
	$= 3.1 \times 10^4$ W	**A1**
1(b)(iv)	(at/after t = 5.8 s) the kinetic energy (of the car) does not change / work is done only against resistive forces / no work is done to accelerate (the car) so (power output is) less	**B1**

Question	Answer	Marks